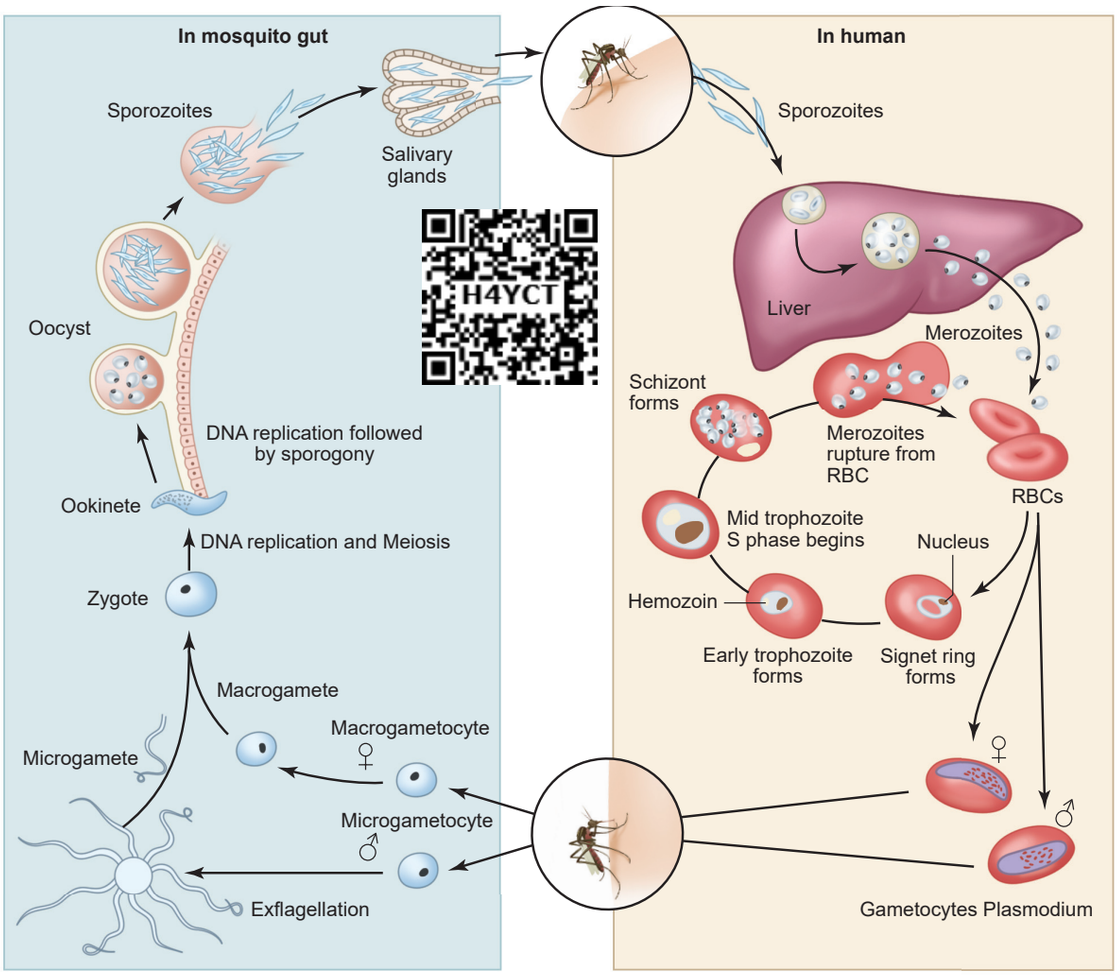


Fig. 7.3 Life cycle of *Plasmodium*

the mosquito. The cycle is now completed and when the mosquito bites another human host, the sporozoites are injected and the cycle begins a new.

The pathological changes caused by malaria, affects not only the erythrocytes but also the spleen and other visceral organs. Incubation period of **malaria** is about 12 days. The early symptoms of malaria are headache, nausea and muscular pain.

The classic symptoms first develop with the synchronized release of merozoites, **haemozoin** toxin and erythrocyte debris into the blood stream resulting in malarial paroxysms – shivering chills, high fever followed by sweating. Fever and chills are caused partly by malarial toxins that induce macrophages to release tumour necrosis factor (TNF-α) and interleukin.

Table 7.3 Types of malaria

Sl. No	Types of Malaria	Causative agent	Duration of Erythrocytic cycle
1	Tertian, benign tertian or vivax malaria	*P. vivax*	48 hours
2	Quartan malaria	*P. malariae*	72 hours
3	Mild tertian malaria	*P. ovale*	48 hours
4	Malignant tertian or quotidian malaria	*P. falciparum*	36 – 48 hours

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